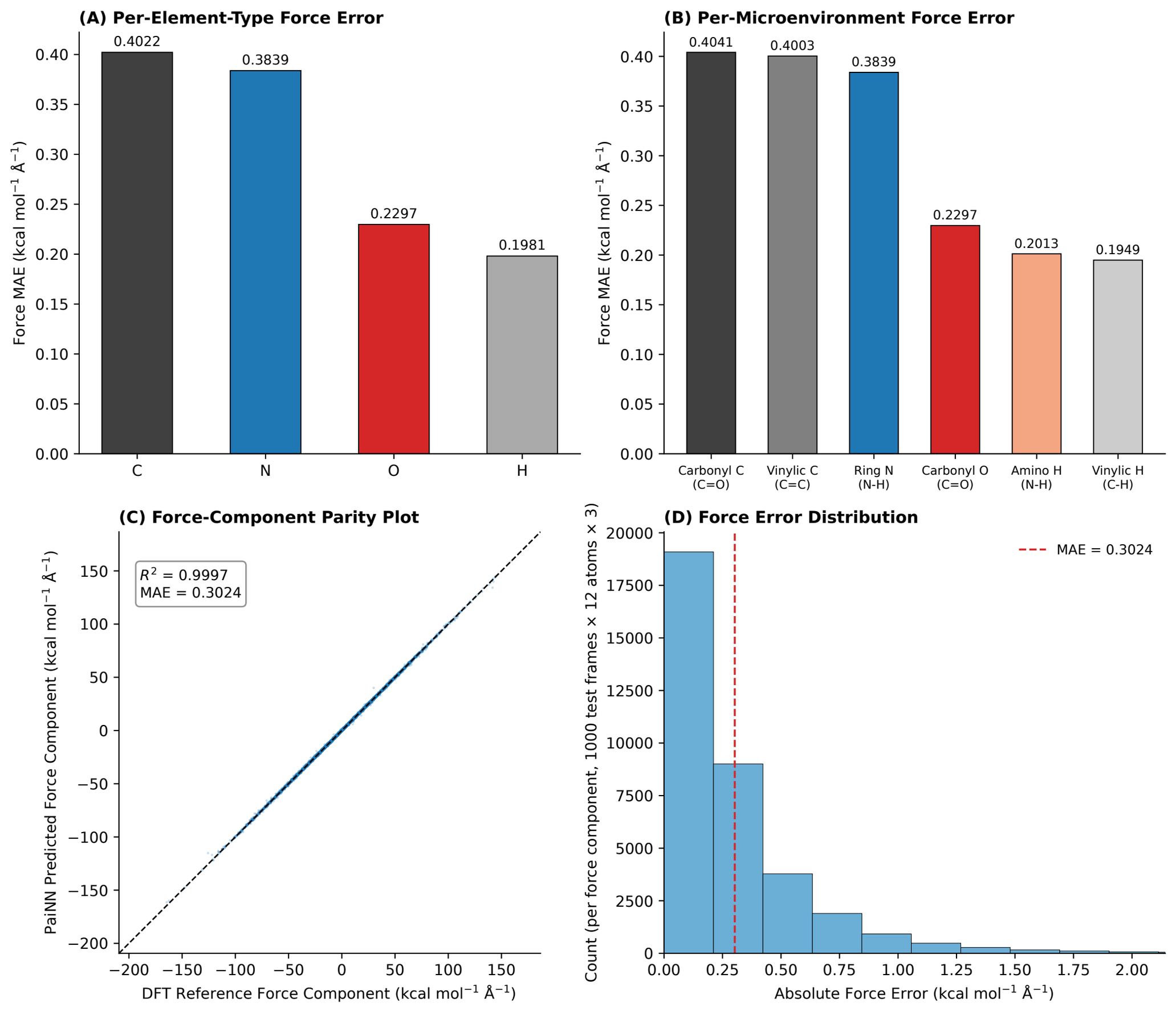

(A) Per-Element-Type Force Error
Force MAE (kcal mol⁻¹ Å⁻¹)
0.4022
0.3839
0.2297
0.1981
C
N
O
H
(B) Per-Microenvironment Force Error
Force MAE (kcal mol⁻¹ Å⁻¹)
0.4041
0.4003
0.3839
0.2297
0.2013
0.1949
Carbonyl C (C=O)
Vinylic C (C=C)
Ring N (N-H)
Carbonyl O (C=O)
Amino H (N-H)
Vinylic H (C-H)
(C) Force-Component Parity Plot
R^2 = 0.9997
MAE = 0.3024
PaiNN Predicted Force Component (kcal mol⁻¹ Å⁻¹)
DFT Reference Force Component (kcal mol⁻¹ Å⁻¹)
(D) Force Error Distribution
MAE = 0.3024
Count (per force component, 1000 test frames × 12 atoms × 3)
Absolute Force Error (kcal mol⁻¹ Å⁻¹)